

# Lecture 17: The Fire Engine problem & Inner Product

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*[Based on slides for „Coding the Matrix“]*



WEEK

NINE



# Agenda

## Chapter 8

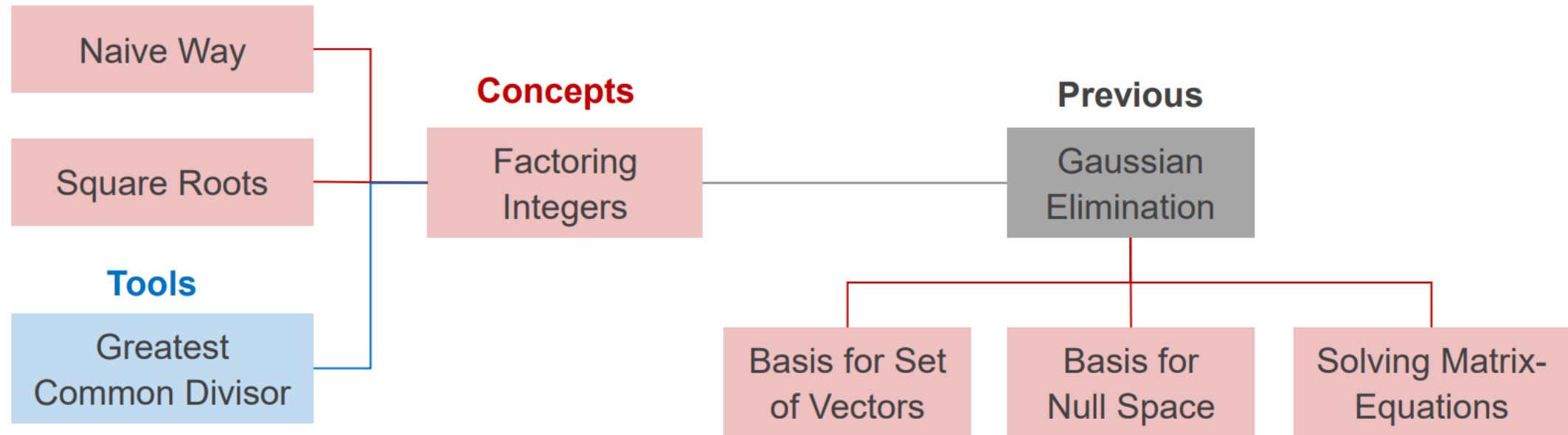
### Lecture 17:

Fire Engine Problem  
Vector norm  
Inner Product  
Orthogonality  
Fire Engine Lemma  
Vector decomposition



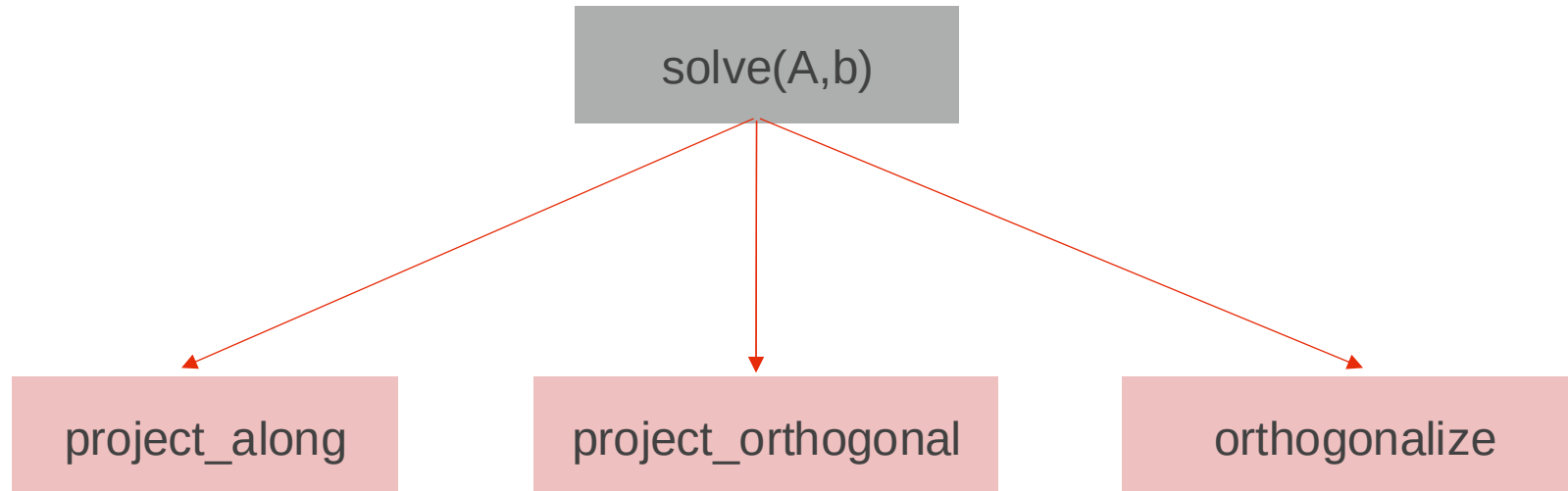


# Lecture 16 - Summary





# Lecture 17(+) - Goal





# Fire Engine Problem

**Problem:** What is the closest path to reach **b** from **v** ?



**Questions:**

**Closest point to b from v ?**

How far is it ?

Let say **p** any vector on **v**, need to define the **length** of **p-b**.



# Distance, length, norm

**Definition 8.1:** For any vector  $\mathbf{v}$ , the length of  $\mathbf{v}$  is now calling the norm and written  $\|\mathbf{v}\|$ .

**Properties 8.1.1 (N1):** For any vector  $\mathbf{v}$ ,  $\|\mathbf{v}\|$  is a nonnegative real number.

**8.1.2 (N2):** For any vector  $\mathbf{v}$ ,  $\|\mathbf{v}\|$  is zero iff  $\mathbf{v}$  is a zero vector.

**8.1.3 (N3):** For any vector  $\mathbf{v}$  and any scalar  $\alpha$ ,  $\|\alpha\mathbf{v}\| = \alpha\|\mathbf{v}\|$ .

**8.1.5 (N4):** For any vector  $\mathbf{v}$  and any vector  $\mathbf{u}$ ,  $\|\mathbf{v} + \mathbf{u}\| \leq \|\mathbf{v}\| + \|\mathbf{u}\|$ .

**Problem:** How calculate the norm of  $\mathbf{v}$  ?



Inner product



# Inner product & norm

**Definition 8.2.1:** The inner product for two vector  $\mathbf{u}$  and  $\mathbf{v}$  over  $\mathbb{R}$  is defined as the dot-product:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \begin{bmatrix} \mathbf{u}^T \end{bmatrix} \begin{bmatrix} \mathbf{v} \end{bmatrix}$$

**Properties 8.2.1.1 (Linearity):**  $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$

**8.2.1.2 (Symmetry):**  $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$

**8.2.1.3 (Homogeneity):**  $\langle \alpha \mathbf{u}, \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle$

**From inner product to norm:** Lets define the norm of  $\mathbf{v}$  by the inner product:

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$



# Norm properties proof

**Proof (N1,N2):** Lets write  $\|\mathbf{v}\| = \|[v_1, \dots, v_n]\| = \sqrt{v_1^2 + \dots + v_n^2} = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$

➡ Trivial.

**Proof (N3):**  $\|\alpha \mathbf{v}\|^2 = \langle \alpha \mathbf{v}, \alpha \mathbf{v} \rangle$   
 $= \alpha \langle \mathbf{v}, \alpha \mathbf{v} \rangle$  By homogeneity  
 $= \alpha \langle \alpha \mathbf{v}, \mathbf{v} \rangle$  By symmetry  
 $= \alpha^2 \langle \mathbf{v}, \mathbf{v} \rangle$  By homogeneity  
 $= \alpha^2 \|\mathbf{v}\|^2$

➡  $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$



# Norm properties proof (optional)

**Proof (N4):**

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \langle u, u + v \rangle + \langle v, u + v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \\ &\leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 \\ &\leq (\|u\| + \|v\|)^2\end{aligned}$$

By linearity

By symmetry & linearity

Using Cauchy-Schwarz  
inequality

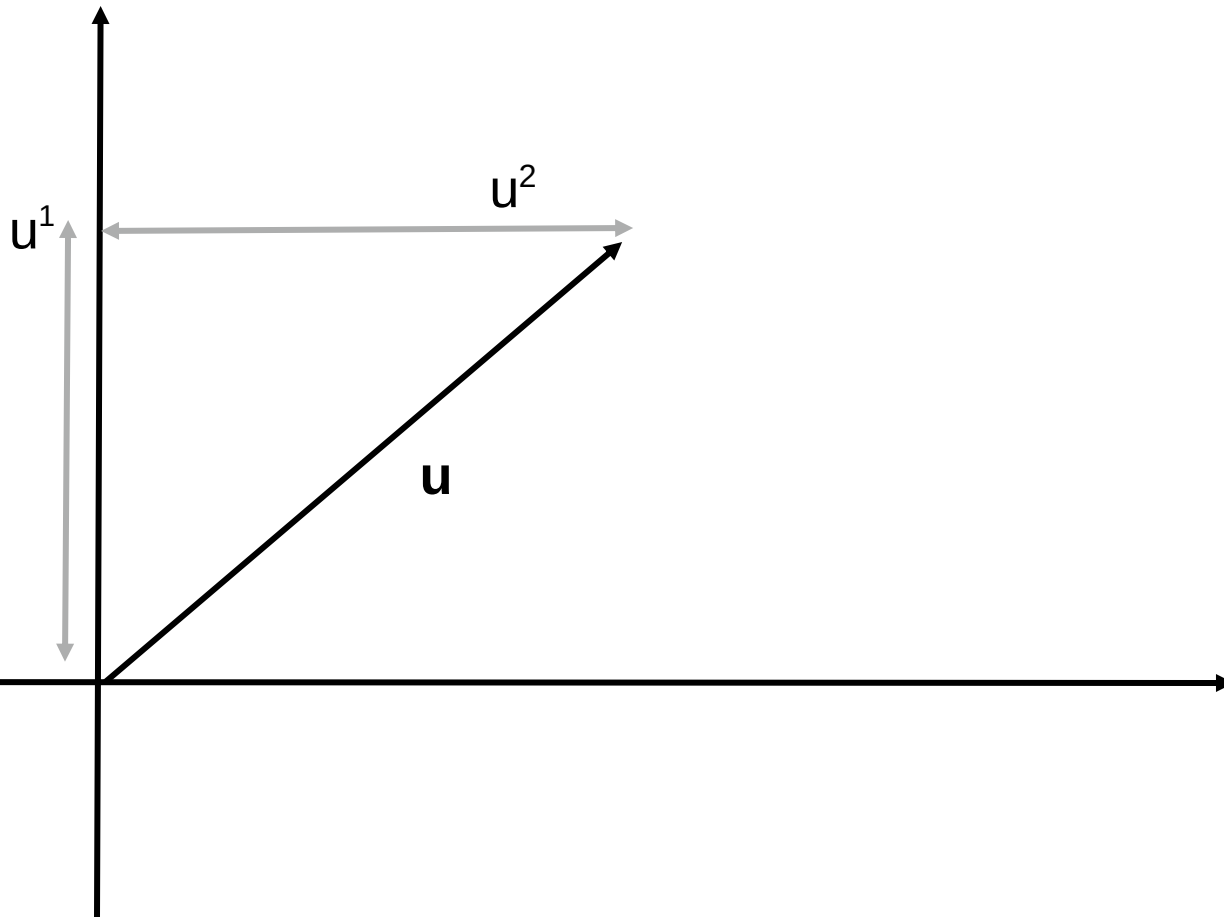
$$\langle u, v \rangle \leq \|u\|\|v\|$$


$$\|u + v\| \leq \|u\| + \|v\|$$



# Example

Lets  $\mathbf{u} := [u_1, u_2]$  , what is  $\|\mathbf{u}\|$ ?



Using Pythagorean theorem:

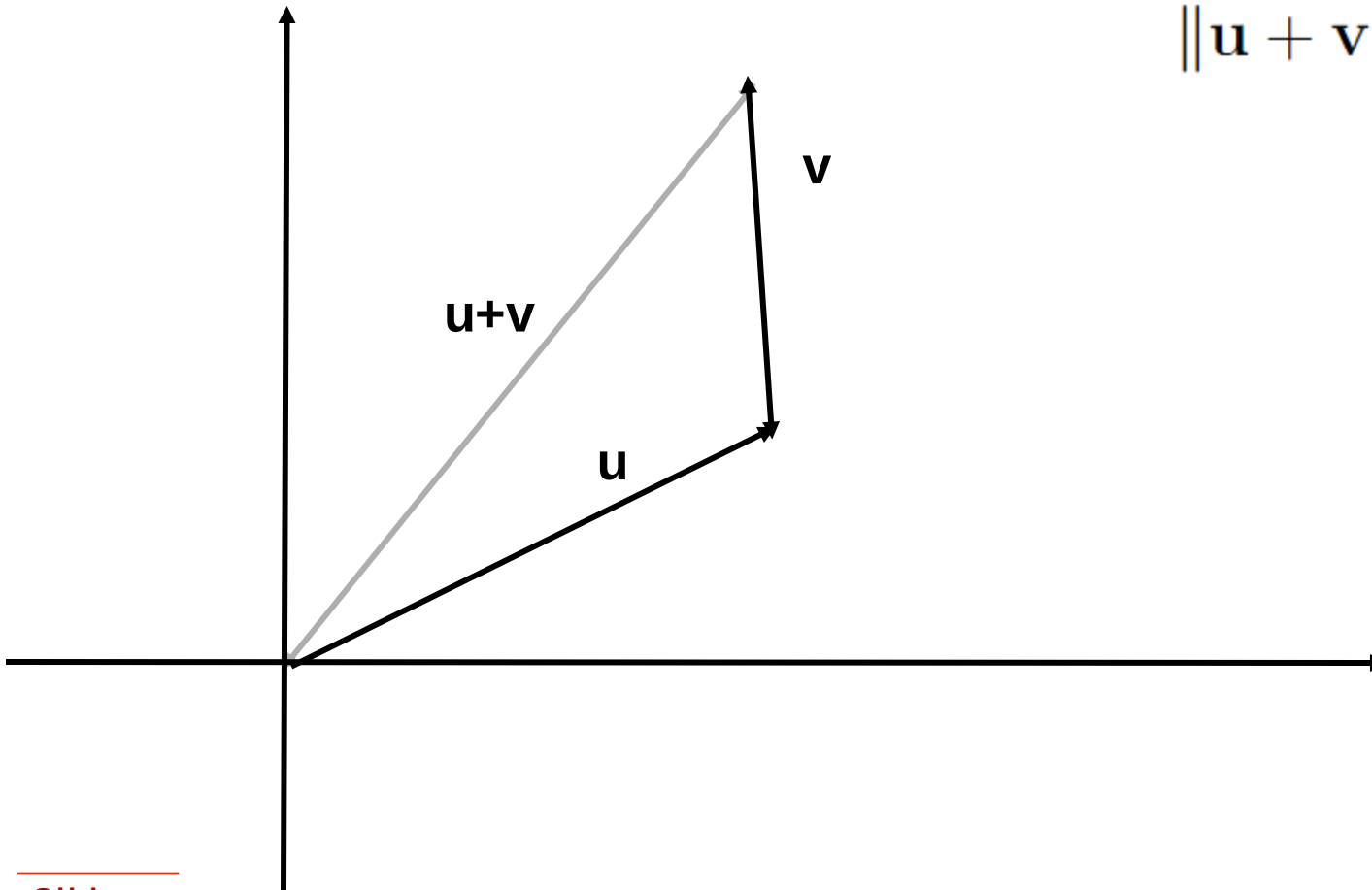
$$\|\mathbf{u}\|^2 := \|u_1\|^2 + \|u_2\|^2$$

Only stand for  $\mathbb{R}^2$  examples.



# Orthogonality (introduction)

Lets  $\mathbf{u}$  and  $\mathbf{v}$  two vectors over the reals, what is the norm of  $\mathbf{u}+\mathbf{v}$  ?



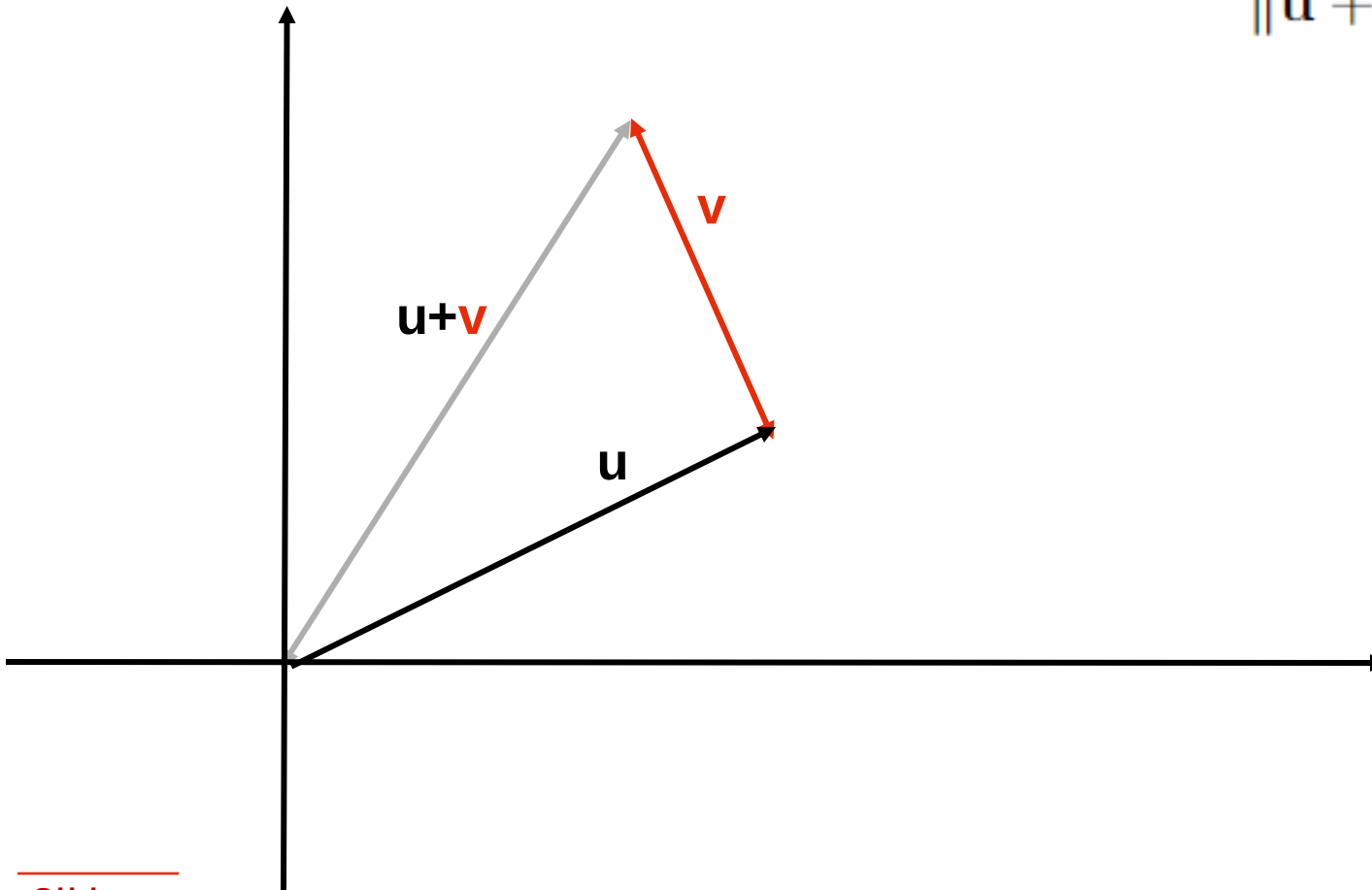
$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2\end{aligned}$$

Cannot go further.



# Orthogonality (introduction)

Lets  $\mathbf{u}$  and  $\mathbf{v}$  two vectors over the reals,  $\mathbf{u}$  is **orthogonal** to  $\mathbf{v}$ , what is the norm of  $\mathbf{u}+\mathbf{v}$  ?



$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\end{aligned}$$



# Orthogonality (definition)

**Definition 8.3:** Let  $\mathbf{u}$  and  $\mathbf{v}$  two real vectors,  $\mathbf{u}$  and  $\mathbf{v}$  are said **orthogonal** if and only if:  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$

**Theorem (Pythagorean theorem for vector over the reals) 8.3.1:** If vectors  $\mathbf{u}$  and  $\mathbf{v}$  over the real are **orthogonal** then:  $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$



# Orthogonality (properties)

**Lemma 8.3.2:** For any vectors  $\mathbf{u}$  and  $\mathbf{v}$  and any scalar  $\alpha$ :

**8.3.2.1 (O1):** if  $\mathbf{u}$  is **orthogonal** to  $\mathbf{v}$  then  $\mathbf{u}$  is orthogonal to  $\alpha\mathbf{v}$ .

**8.3.2.1 (O2):** if  $\mathbf{u}$  and  $\mathbf{v}$  are both **orthogonal** to  $\mathbf{w}$  then  $\mathbf{u}+\mathbf{v}$  is orthogonal to  $\mathbf{w}$ .

**Proof:**

1.  $\langle \mathbf{u}, \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle = \alpha 0 = 0$
2.  $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle = 0 + 0$



# Orthogonality (Examples)

**Example:**  $[1, 2] \cdot [2, -1] = 0$  so  $[1, 2] \cdot [20, -10] = 0$

**Example:**

$$\begin{array}{rcl} [1, 2, 1] \cdot [1, -1, 1] & = & 0 \\ [0, 1, 1] \cdot [1, -1, 1] & = & 0 \\ \hline ([1, 2, 1] + [0, 1, 1]) \cdot [1, -1, 1] & = & 0 \end{array}$$



# Orthogonality & fire engine problem

**Fire Engine Lemma (intro):** Let  $\mathbf{b}$  be a vector and  $\mathbf{a}$  a nonzero vector such the set  $\{\alpha\mathbf{a}: \alpha \text{ any real}\}$  is a line  $\mathbf{L}$ . Let define  $p$  be the point on the line  $\mathbf{L}$  such  $\mathbf{b}-\mathbf{p}$  is **orthogonal** to

$\mathbf{a}$ .

Then  $p$  is the point on  $\mathbf{L}$  **closest** to  $\mathbf{b}$ .

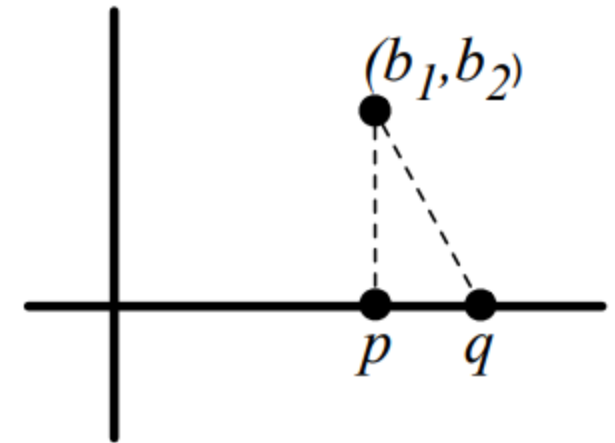
**Example in  $\mathbb{R}^2$ :** Let the line  $\mathbf{L}$  being the x-axis define as the set  $\{(x,y): y=0\}$  and the point  $b$  is  $(b_1, b_2)$ .

From Lemma:  $p=(b_1, 0)$  **closest** to  $b$ .

For any other point  $q$  on  $\mathbf{L}$  different to  $p$ :  $p, q$  and  $b$  build **a right triangle**.

Using **Pythagorean theorem** norm of  $\mathbf{b}-q$  is always greater than the height.

Shows that  $q$  is farther from  $b$  than  $p$  is.





# Orthogonality & fire engine problem

**Proof:** Let  $\mathbf{q}$  be any point of  $\mathbf{L}$  different of  $\mathbf{p}$ .

Since  $\mathbf{b}-\mathbf{p}$  is orthogonal to  $\mathbf{a}$ ,  $\mathbf{p}-\mathbf{q}$  being on the line  $\mathbf{L}$  is also orthogonal to  $\mathbf{b}-\mathbf{p}$ .

➡ The points  $\mathbf{b}, \mathbf{p}$  and  $\mathbf{q}$  form a right triangle.

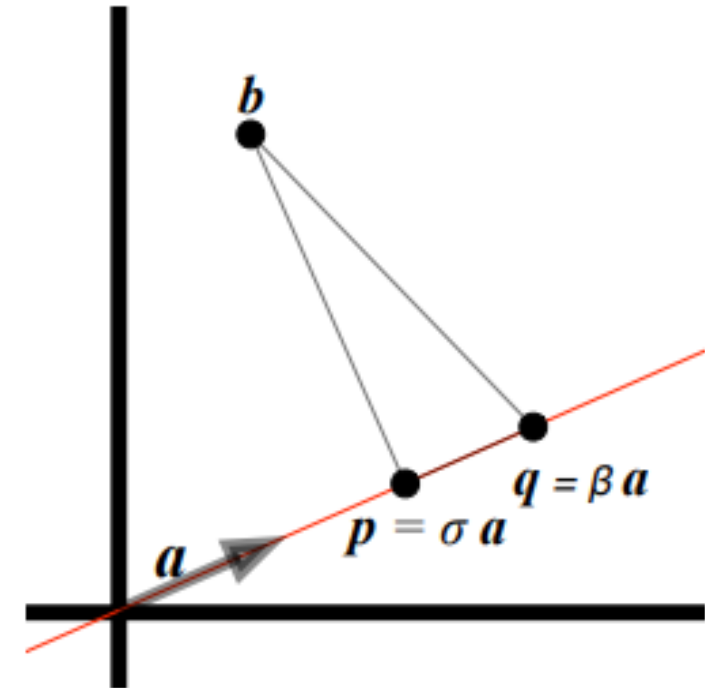
Using **Pythagorean theorem** norm of  $\mathbf{b}-\mathbf{q}$ :

$$\|\mathbf{b} - \mathbf{q}\|^2 = \|\mathbf{p} - \mathbf{q}\|^2 + \|\mathbf{b} - \mathbf{p}\|^2$$

➡ 
$$\|\mathbf{b} - \mathbf{p}\|^2 = \|\mathbf{b} - \mathbf{q}\|^2 - \|\mathbf{p} - \mathbf{q}\|^2$$

For any  $\mathbf{q} \neq \mathbf{p}$ ,  $\|\mathbf{p} - \mathbf{q}\|^2 > 0$

➡ 
$$\|\mathbf{b} - \mathbf{p}\|^2 < \|\mathbf{b} - \mathbf{q}\|^2$$





# Decomposition of $\mathbf{b}$

**Definition 8.3.6:** For any vector  $\mathbf{b}$  and any vector  $\mathbf{a}$ , define vectors  $\mathbf{b}^{\parallel \mathbf{a}}$  and  $\mathbf{b}^{\perp \mathbf{a}}$  to be the *projection of  $\mathbf{b}$  onto  $\text{Span}\{\mathbf{a}\}$*  and the *projection of  $\mathbf{b}$  orthogonal to  $\mathbf{a}$*  if

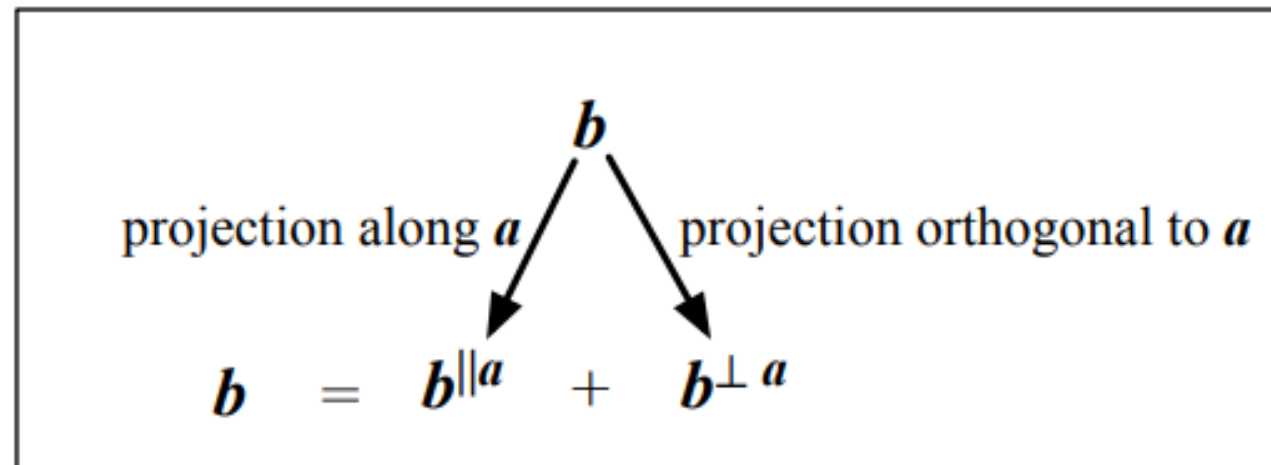
$$\mathbf{b} = \mathbf{b}^{\parallel \mathbf{a}} + \mathbf{b}^{\perp \mathbf{a}}$$

and there is a scalar  $\sigma \in R$  such that

$$\mathbf{b}^{\parallel \mathbf{a}} = \sigma \mathbf{a}$$

and

$\mathbf{b}^{\perp \mathbf{a}}$  is orthogonal to  $\mathbf{a}$





# Closest-Point Corollary & Example

**Fire Engine Lemma (8.3.8):** For any  $\mathbf{b}$  and vector  $\mathbf{a}$  over the reals,

→ The projection  $\mathbf{b}^{\parallel \mathbf{a}}$  of  $\mathbf{b}$  onto  $\text{Span}\{\mathbf{a}\}$  is the **closest** to  $\mathbf{b}$ .

→ The distance between  $\mathbf{b}$  and  $\mathbf{b}^{\parallel \mathbf{a}}$  is  $\|\mathbf{b}^{\perp \mathbf{a}}\|$

**Example:** What if  $\mathbf{a}$  is the zero vector ? Show that the closest point to  $\mathbf{b}$  is

$\mathbf{b}$ . → The projection  $\mathbf{b}^{\parallel \mathbf{a}} = 0$ , according to the previous definition  $\mathbf{b} = \mathbf{b}^{\perp \mathbf{a}}$

→  $\mathbf{b}^{\perp \mathbf{a}}$  is **orthogonal** to  $\mathbf{a}$ , since every vectors are **orthogonal** to the zero vector  $\mathbf{a}$ .

→ The only point in  $\text{Span}\{\mathbf{a}\}$  is the zero vector, the **closest** point to  $\mathbf{b}$  is the zero vector and the distance is  $\|\mathbf{b}\|$ .



# Computing the projections (Analysis)

**Problem:** Give an expression for the projection of  $\mathbf{b}$  onto  $\mathbf{a}$ :  $\mathbf{b}^{\parallel \mathbf{a}}$

**Use:**  $\mathbf{b} = \mathbf{b}^{\parallel \mathbf{a}} + \mathbf{b}^{\perp \mathbf{a}}$

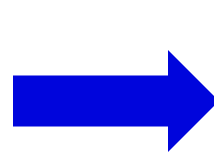
$$\mathbf{b}^{\parallel \mathbf{a}} = \sigma \mathbf{a}$$

$\mathbf{b}^{\perp \mathbf{a}}$  is orthogonal to  $\mathbf{a}$

 If  $\mathbf{a} = \mathbf{0}$ , then  $\mathbf{b}^{\parallel \mathbf{a}} = \mathbf{0}$ 
 If  $\mathbf{a} \neq \mathbf{0}$ , then  $\mathbf{b}^{\parallel \mathbf{a}} = \sigma \mathbf{a}$  Need to express **sigma** !

  $\langle \mathbf{b}^{\perp \mathbf{a}}, \mathbf{a} \rangle = 0$ . Substitute for  $\mathbf{b}^{\perp \mathbf{a}}$ :  $\langle \mathbf{b} - \mathbf{b}^{\parallel \mathbf{a}}, \mathbf{a} \rangle = 0$ .

 Using **linearity** and **homogeneity** of the inner product  $\langle \mathbf{b}, \mathbf{a} \rangle - \sigma \langle \mathbf{a}, \mathbf{a} \rangle = 0$

  $\sigma = \frac{\langle \mathbf{b}, \mathbf{a} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle}$  **Lemma (8.3.11)**

In the special case in which  $\|\mathbf{a}\| = 1$ , the denominator  $\langle \mathbf{a}, \mathbf{a} \rangle = 1$  so

$$\sigma = \langle \mathbf{b}, \mathbf{a} \rangle$$



# Computing the projections (Computational)

**Problem:** Write a code to compute the projection of **b** onto **a**:  $b^{\parallel a}$

**Use:**  $b^{\parallel a} = \sigma a \quad \sigma = \frac{\langle b, a \rangle}{\langle a, a \rangle}$

**Answer:** `def project_along(b, a): return ((b*a)/(a*a))*a` **Almost.**

**Best:**

`def project_along(b, a): return ((b*a)/(a*a) if a*a != 0 else 0)*a`

**Revised version:**

```
def project_along(b, a):
    sigma = (b*a)/(a*a) if a*a > 1e-20 else 0
    return sigma * a
```



# Solution to the fire engine problem

**Problem:** What is the closest path to reach  $\mathbf{b} = [2, 4]$  from  $\mathbf{a} = [6, 2]$  ?

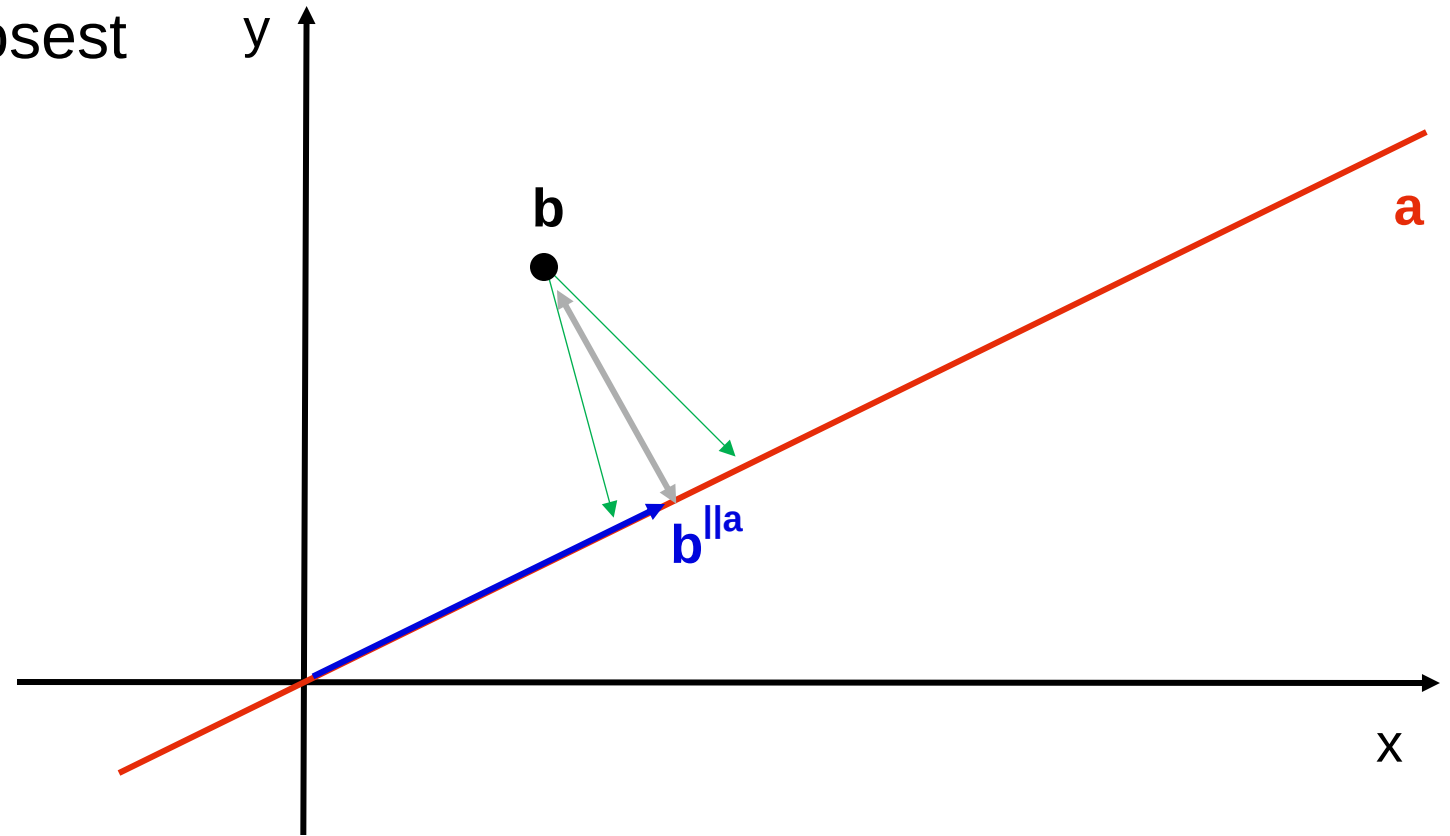
➔ **Fire engine Lemma:** The closest point from  $\mathbf{b}$  to  $\mathbf{a}$  is  $\mathbf{b}^{\parallel \mathbf{a}} = \sigma \mathbf{a}$

$$\begin{aligned} \sigma &= \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} \\ &= \frac{6 \cdot 2 + 2 \cdot 4}{6 \cdot 6 + 2 \cdot 2} \\ &= \frac{20}{40} \\ &= \frac{1}{2} \end{aligned}$$

➔  $\mathbf{b}^{\parallel \mathbf{a}} = 1/2[6, 2] = [3, 1]$

The distance to  $\mathbf{b}$  is

$$\|\mathbf{b}^{\perp \mathbf{a}}\| = \|[2, 4] - [3, 1]\| = \|[-1, 3]\| = \sqrt{10}$$





# Lecture 17 - Recap

## Theorems & Lemmas

Fire Engine Lemma

Projection of  
b along a

Project\_along

Pythagorean theorem

## Concept & Definition

Inner Product

Vector Norm

Orthogonality

## Previous

Solving Matrix  
vector  
equation

Tools

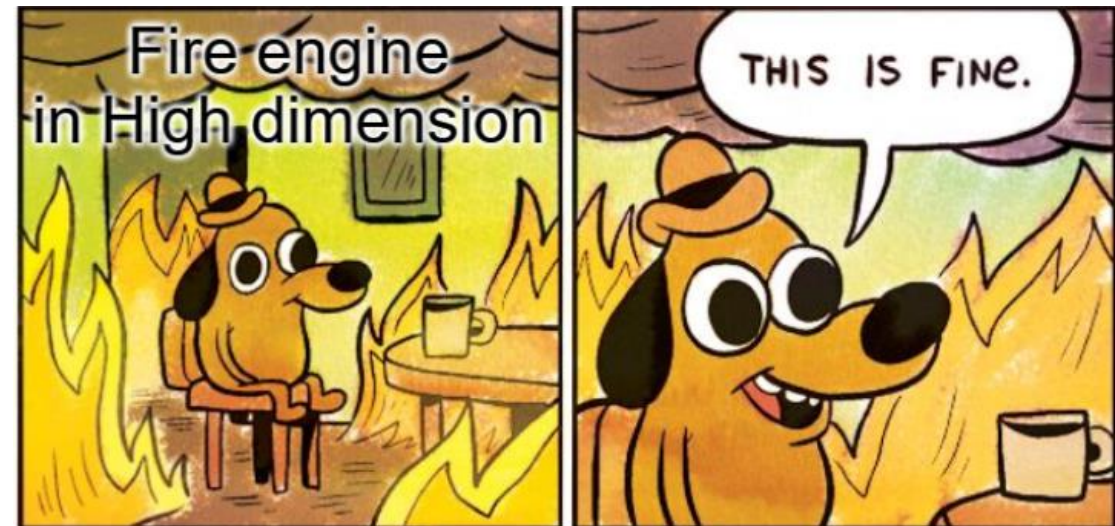


# Towards solving the higher dimensional version of best approximation

A natural generalization of the *fire engine* problem is this:

**Computational Problem:** *Closest point in the span of several vectors*

Given a vector  $\mathbf{b}$  and vectors  $\mathbf{a}_1, \dots, \mathbf{a}_n$  over the reals, find the vector in  $\text{Span} \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$  closest to  $\mathbf{b}$ .





# Thank You For Your Attention!





# APPENDIX



# Applied exercise



# Theoretical exercise